Design to Cost · AUDIT · TOOL DESIGN, DRAWINGS, SIMULATION, 3D CORRECTION

Is every task complete for every type of mould and tool?

Audit of release 2026-10-26 from a tool-room head's point of view. 42 of 63 task × tool-type cells are done; sheet-metal dies and injection moulds are complete; die-casting, EPS and blow moulds are complete for the mould structure with their process-specific items listed for the designer; foundry is not covered yet. Every output was made from the sample parts and checked (pictures, drawings, STEP files, automated tests).

Task	Sheet metal: multi-stage dies	Sheet metal: progressive die	Plastic injection mould	Die-casting die	EPS mould	Stretch-blow mould	Foundry pattern / core box
3D assembly (STEP) with every part	**Done** Blanking, piercing, trimming, V-bending: shoes, pillars, bushes, die plate, backing plates, guides / nest, stop pin, stripper, punch holder, punches, screws, dowels	**Partly** Piercing + cut-off stations at the pitch; forming stations, pilots and lifters are left to the designer	**Done** Core / cavity from the part × shrinkage, A / B / clamp / support plates, spacer blocks, ejector plates and pins, return pins, pillars, bushes, sprue bush, locating ring, cooling	**Partly** Same structure with die-casting rules; shot sleeve, runner, overflows and vacuum vents are left to the designer	**Partly** Cavity and core halves in the steam-chest base; fill injectors, vents and steam chambers are left to the designer	**Partly** Two cavity halves split through the middle; neck / base inserts and blow pin are left to the designer	**Not yet** No tool cost for foundry yet (planned stage), so no tool design
Every part as its own 3D (STEP)	**Done**	**Done**	**Done**	**Done**	**Done**	**Done**	**Not yet**
2D drawings: assembly + detail of every made part (PDF and DXF)	**Done** Assembly: iso, exploded, section, BOM. Plates: true views, hole table with fits and depths. Punches / V-die: three views; profiles in DXF for wire cut	**Done** As multi-stage	**Done** Plates as above; core / cavity: three views + STEP (3D-machined)	**Done** As plastic	**Done** As plastic	**Done** As plastic	**Not yet**
Tolerances, fits, finish, heat treatment, labels on drawings	**Done** General tolerance ISO 2768-mK, plate thickness / flatness / parallelism, fits (pillar g6, bush H7, dowel and punch m6/H7), position of fitted holes, finishes, hardness, heat-treatment note - all Design rules	**Done**	**Done** + insert-in-pocket fit, cavity polish	**Done**	**Done**	**Done**	**Not yet**
Logo and title block on every drawing (CAD and PDF)	**Done** Admin → Drawings: logo, company, projection, drawn / checked / approved, notes; DXF logo as hatches	**Done**	**Done**	**Done**	**Done**	**Done**	**Not yet**
Simulation report (PDF)	**Done** Forming report: checks OK / FAIL, blank and yield, stages, force, press, pictures per stage, flat pattern	**Done** Same report	**Done** Wall / draft / sink / undercut maps, cycle, clamp, machine	**Done** Same analysis	**Not yet** No simulation rules for EPS yet	**Not yet** No simulation rules for blow moulding yet	**Not yet**
Simulation video (MP4)	**Done** Blanking, piercing, bends folding; failing bend in red	**Done**	**Done** Turntable of the wall, draft and risk maps with the result	**Done**	**Not yet**	**Not yet**	**Not yet**
Correct the 3D model in the software and release it as a new revision	**Done** Bend radius (outside size kept) and small holes; preview, STEP, new revision	**Done** Same	**Not yet** Draft, wall and rib changes need CAD: upload the new revision	**Not yet** As plastic	**Not yet**	**Not yet**	**Not yet**

Task	Sheet metal: multi-stage dies	Sheet metal: progressive die	Plastic injection mould	Die-casting die	EPS mould	Stretch-blow mould	Foundry pattern / core box
Release pack ZIP for the tool room (3D + PDF + DXF + BOM)	**Done**	**Done**	**Done**	**Done**	**Done**	**Done**	**Not yet**

Checked and right

- Blanking: die opening = blank size, punch = blank – clearance; piercing: punch = hole, die = hole + clearance (clearance % of t per side, Design rules).
- Flat pattern unfolded from the 3D model: the sample L bracket corrected in the software to R2 unfolds to exactly the designer's Rev B (326.178 mm).
- Bend correction keeps the outside size (checked on rounded and sharp-cornered models) and the holes; the new revision passes the forming check.
- Die plate and punches D2 58-62 HRC, shoes and plates from the die set / mould base material, grades from the tool BOM (Grade rules, your choices).
- Every plate has its holes in a table from the datum corner: through holes, counterbores, tapped holes, fits with position tolerance.
- Mould: core and cavity cut from the part with shrinkage; ejector pins cut to the core surface; cooling channels kept clear of the part.
- No number is fixed in the program: sizes, fits, tolerances and finishes are Design rules (EXAMPLE values flagged until you replace them).

Check before release (tool designer)

- Press shut height and bed size against the press data sheet (shut height is on the assembly sheet).
- Stripper: the design uses a fixed stripper on stock guides / nest; use a spring stripper where flatness of the part needs it.
- V-bending: pressure pad, back gauge position and springback compensation by try-out.
- Moulds: gate, runner and vents; slides and lifters where the concept analysis found undercuts; hot runner if chosen in the tool cost.
- Detail drawings of punches, V-die and inserts give the overall sizes; profiles come from the DXF / STEP (wire cut, CNC).

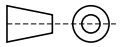
Not done yet (next stages)

- Foundry: pattern, core boxes and moulding boxes (no tool cost yet).
- Drawing dies (deep draw) and the forming stations of progressive dies in 3D.
- Simulation and video for EPS and blow moulding.
- 3D correction of draft, wall thickness and ribs (needs CAD).

Notes: 1. Remove all sharp edges 0.2 x 45°. Do not scale the drawing. The 3D model is the master.

Design to Cost
Drawing no. **DTC-AUDIT-27**
Rev **-**

Scope	Roadmap stage 27	**Audited**	2026-10-26

General tolerance	ISO 2768-mK
Projection	
Scale	-
Sheet	1 / 1
Drawn	- · 2026-10-26
Checked	-
Approved	-